Machine Learning - Undergraduate Level Assessment 2 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Suppose we like to calculate P(H|E, F) and we have no conditional independence information. Which of the following sets of numbers are sufficient for the calculation?
   (a) P(E, F), P(H), P(E|H), P(F|H)
   (b) P(E, F), P(H), P(E, F|H)
   (c) P(H), P(E|H), P(F|H)
   (d) P(E, F), P(E|H), P(F|H)

2. Which of the following can only be used when training data are linearly separable?
   (a) Linear hard-margin SVM.
   (b) Linear Logistic Regression.
   (c) Linear Soft margin SVM.
   (d) The centroid method.

3. Predicting the amount of rainfall in a region based on various cues is a _________ problem.
   (a) Supervised learning
   (b) Unsupervised learning
   (c) Clustering
   (d) None of the above

4. K-fold cross-validation is
   (a) linear in K
   (b) quadratic in K
   (c) cubic in K
   (d) exponential in K

5. Which of the following sentence is FALSE regarding regression?
   (a) It relates inputs to outputs.
   (b) It is used for prediction.
   (c) It may be used for interpretation.
   (d) It discovers causal relationships

True / False (10 marks)

Answer True or False

6. True or False: Batch Normalization is used in the original ResNet paper, while Layer Normalization is not.
7. True or False: The classification run time of the nearest neighbors algorithm is O(log N), indicating a more efficient search process.
8. True or False: Using strong classifiers is effective in preventing overfitting when performing bagging.
9. True or False: A Bayesian network with structure H -> U <- P <- W requires only 4 independent parameters for full specification.
10. True or False: ImageNet contains images of various resolutions, providing a diverse dataset for image classification tasks.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find out the minimum no of swaps required for bracket balancing in the given string.
12. Write a function to select the nth items of a list.

End of Paper